

1) First Derivative: Rate of Reaction

The first derivative shows how the concentration of a reactant or product changes over time. This is most common derivative in chemical kinetics.

Ex: For a reaction $A \rightarrow B$, the concentration of A decreases over time.

The reaction is: $\text{Rate} = -\frac{d[A]}{dt}$

If the reaction is first-order, the rate law is:

$$\text{Rate} = k[A]$$

The concentration of A at time t can be written as:

$$[A] = [A]_0 e^{-kt} \quad \text{Here, } [A]_0 = \text{Initial concentration}$$

$$\text{1st derivative rate: } -\frac{d[A]}{dt} = k[A]$$

2) Second Derivative: Change in Rate (Acceleration)

The second derivative shows how the rate of reaction changes over time. It's the derivative of the rate.

Ex: Second Derivative in Concentration change:

For a first-order reaction $A \rightarrow B$:

From this equation: $\frac{d[A]}{dt} = -k[A]_0 e^{-kt}$

The second derivative with respect of time is: $\frac{d^2[A]}{dt^2} = k^2[A]_0 e^{-kt}$

Meaning: The second derivative shows that the rate is decreasing over time (Negative slope) because A is consumed. The curvature indicates the acceleration of the rate decrease.

3) Third Derivative: Rate of Change of Acceleration
 The 3rd derivative is the derivative of the second derivative. It tells how the acceleration of the rate changes over time.

Ex: Third Derivative in kinetics

For the same first-order reaction $A \rightarrow B$.

Starting from: $\frac{d[A]}{dt} = -k[A]_0 e^{-kt}$

The third derivative is: $\frac{d^3[A]}{dt^3} = -k^3[A]_0 e^{-kt}$

Meaning: The third derivative tells us that the rate of change of acceleration is constant for this exponential decay process.

Key Note:

(i) First derivative: Most common in reaction rate calculations.

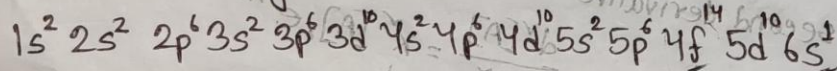
(ii) Second derivative: Useful of study how the rate changes over time.

(iii) Third derivative: Rare in practical chemistry but shows deeper changes in kinetics.

Gold, 79 e⁻ विभाग:

Gold has 79 electrons arranged in 8 different orbitals:

It's electronic configuration is:



Bond Energy: Bond energy is defined as the energy required to split the molecule into atoms. It depends on types of bonding - single, double or triple bonds.

Chemical Bond
Defination: A chemical bond is the force of attraction that holds atoms or ions together in a molecule or compound.

Types of Chemical Bonds:

1. Ionic Bond: The bond formed when electrons are transferred from one atom to another, creating oppositely charged ions that attract each other.

Example: Na^+ and Cl^- from NaCl .

2. Covalent Bond: Formed when atoms share one or more pairs of electrons to achieve stability.

Example: $\text{H} + \text{O}_2 \rightarrow \text{H}_2\text{O}$ (water); H_2O involves shared electrons between hydrogen and oxygen.

3. Hydrogen Bond: A weak bond formed between a hydrogen atom and another electronegative atom.

Example: The bonds between water molecules.

4. Van der Waals Forces: Weak attractions between molecules or atoms due to temporary dipoles.

5. Coordinate Covalent Bond: A type of covalent bond in which both shared electrons in the bond come from the same atom.

Example: When ammonia (NH_3) reacts with a proton (H^+) to form an ammonium ion (NH_4^+): $\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$

The lone pair is donated to the H^+ , forming a coordinate covalent bond.

6. Metallie Bond: Found in metals, where valence electrons are shared in a "sea of electrons" that moves freely around metal ions.

Example: Metallie bonding in Cu (copper)

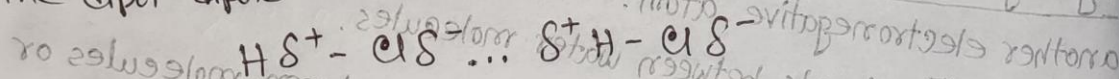
Valence Electron: The electrons in the outer most energy level in an atom that takes part in chemical bonding are called valence electrons.

Bonding electron: The valence electrons actually involved in bond formation are called bonding electrons.

Octet Rule: The tendency for atoms to have eight electrons in the outer shell is known as Octet rule.

Dipole-Dipole Forces: A type of intermolecular force that occurs between polar molecules. These forces arise due to the attraction between the positive end of one polar molecule and the negative end of another polar molecule.

The dipole-dipole interaction occurs as follows:



The positive end (H) of one HCl molecule is attracted to the negative end (Cl) of another HCl molecule.

London Dispersion Forces: Are weak intermolecular forces that arise due to temporary fluctuations in the electron distribution of atoms or molecule.

Bond length: Bond length is the average distance between the nuclei of two bonded atoms in a molecule.

Bond Angle: The angle between the directions of two bonds in a molecule is called the 'bond angle'.

Bond angle depends on:
(i) Charge distribution
(ii) Geometry of the molecules
(iii) Symmetry
(iv) Hybridization

Chemistry

Characteristics of different types of Bonds.

① Ionic Bond:

- ① Strong electrostatic attraction between ions.
- ② High melting and boiling points.
- ③ Soluble in water; conduct electricity in molten or aqueous state.

Example: NaCl (Sodium chloride)

② Covalent Bond:

- ① Strong bonds within molecules.
- ② Can be polar or non polar.
- ③ Low melting and boiling points.
- ④ Poor conductors of electricity.

Example: H_2O

③ Metallic Bond:

- ① Strong bond within metals.
- ② Good conductors of heat and electricity.
- ③ Malleable and ductile.
- ④ Lustrous appearance.

Example: Cu .

④ Hydrogen Bond

- ① Weaker than covalent bonds but stronger than van der Waals force
- ② Responsible for high boiling points in substance like water
- ③ important in DNA structure and protein folding

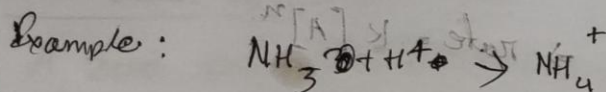
Example: ~~Net~~ Between water molecules H_2O (H_2O)

⑤ Van der Waals forces:

- ① Weakest types of attraction
 - ② increases with molecular size and surface area
 - ③ Present in molecules but significant in nonpolar substances
- Example: Attraction in noble gases like (Ar) (argon).

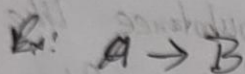
⑥ Coordinate Covalent Bond

- ① Similar strength to regular covalent bonds.
- ② Often forms between a lone pair donor and an electron deficient acceptor.
- ③ Common in complex ions and ligands.



Chemical kinetics

The branch of Physical Chemistry which deals with the rate reactions is called Chemical kinetics



$$\begin{aligned}\text{The rate of the reaction} &= - \frac{d[A]}{dt} \quad [\text{of } A] \\ &= + \frac{d[B]}{dt} \quad [\text{of } B]\end{aligned}$$

Units of rate:

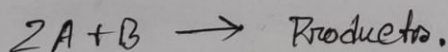
$$\begin{aligned}\text{mole/litre sec} &\text{ or } \text{mol l}^{-1} \text{ s}^{-1} \\ \text{mole/litre min} &\text{ or } \text{mol l}^{-1} \text{ min}^{-1} \\ \text{mole/litre hour} &\text{ or } \text{mol l}^{-1} \text{ h}^{-1}\end{aligned}$$

Rate Laws:

The rate of a reaction is directly proportional to the reactant concentrations, each ~~own~~ raised to a power

$$\begin{aligned}\text{Rate} &\propto [A]^n \\ \Rightarrow \text{rate} &= k [A]^n\end{aligned}$$

For a Reaction:



$$\text{rate} = k [A]^m [B]^n$$

Molecularity of a Reaction

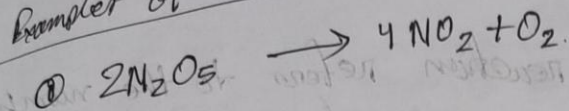
molecularity of a reaction refers to the number of molecular atoms participating in an elementary step.

Chemical reactions are of two types.

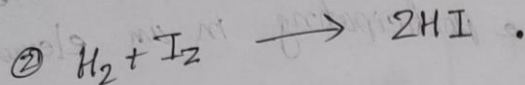
① Elementary Reaction ; it occurs in a single step.

② Complex reaction ; it occurs in two or more steps.

Examples of Rate laws:



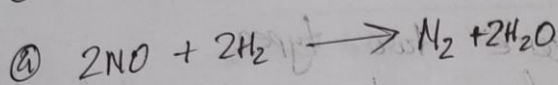
$$\text{rate} = k[\text{N}_2\text{O}_5]$$



$$\text{rate} = k[\text{H}_2][\text{I}_2]$$



$$\text{rate} = k[\text{NO}_2]^2$$



$$\text{rate} = k[\text{H}_2][\text{NO}]^2$$

Order of Reaction:

The order of a reaction is defined as the sum of the powers of concentrations in the rate law.

$$\text{rate} = k[\text{A}]^m[\text{B}]^n ; \text{Order of reaction is } (m+n)$$

Example: Rate law
 $\text{rate} = k[\text{N}_2\text{O}_5]$

Reaction order

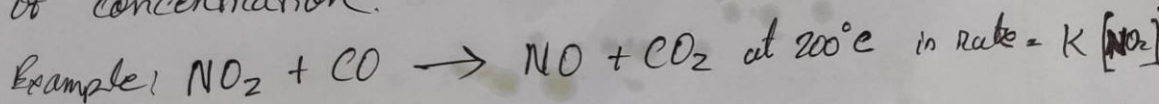
1.

$$\text{rate} = k[\text{H}_2][\text{I}_2]$$

2.

Zero order Reaction.

A zero order reaction is one where rate is independent of concentration.



Order of a Reaction	Molecularity of a Reaction
① it is the sum of powers of the concentrations terms in the law of mass action	① It is number of reacting species undergoing simultaneous collision in the elementary or simple reaction
② It is an experimentally determined value	② It is a theoretical concept
③ It can have fractional value	③ It is always a whole number
④ It can assume zero value	④ It can not have zero value
⑤ Order of a reaction can change with the conditions such as pressure, temperature, concentration	⑤ Molecularity is invariant for a chemical equation

Arrhenius Equation

Where

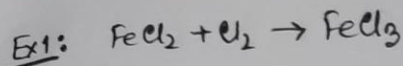
Logarithm form of Arrhenius Equation.

$$\ln K = \ln A e^{-E_a/RT} = \ln A - \frac{E_a}{R} \cdot \frac{1}{T}$$

$$\log K = \frac{-E_a}{2.303 RT} + \log A$$

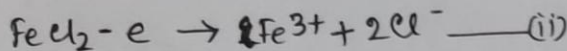
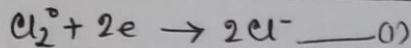
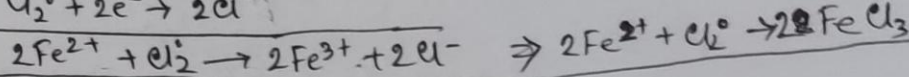
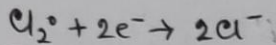
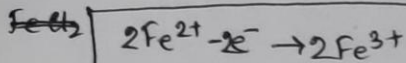
$$\left[R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1} \right]$$

$$e \approx 2.718$$

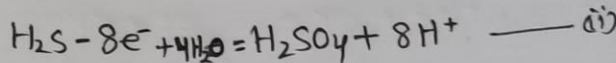
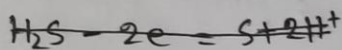
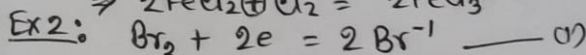
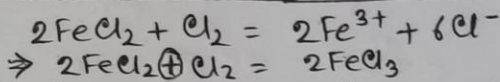
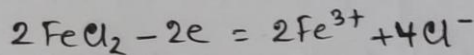
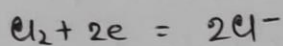


Cl- ज्ञात (ग्रह)

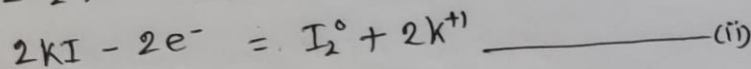
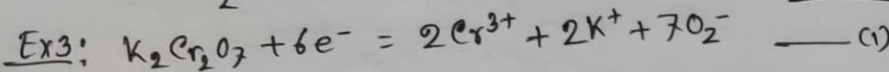
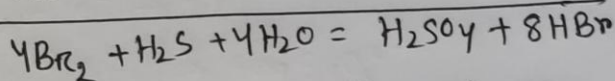
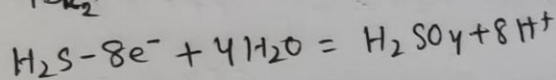
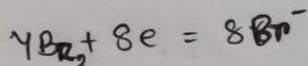
FeCl_2 - लिख (ग्रह)



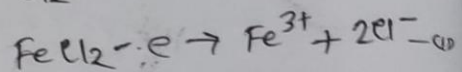
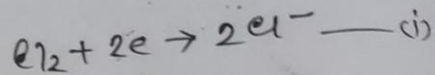
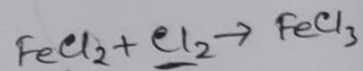
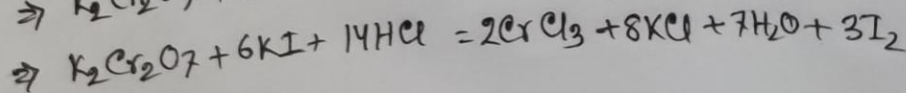
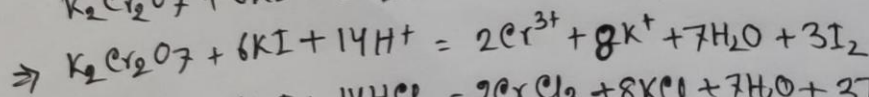
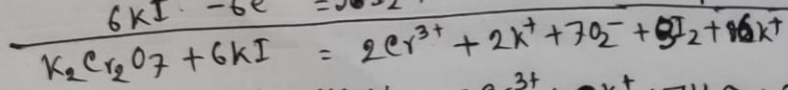
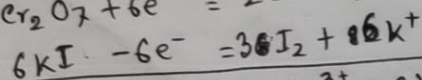
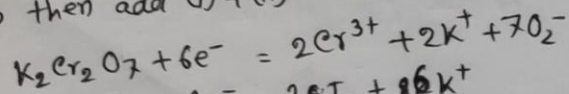
(ii) x 2 for equal the electron number,



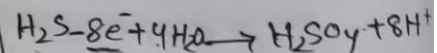
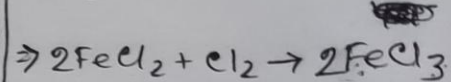
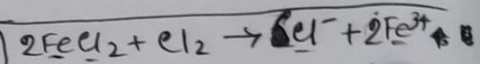
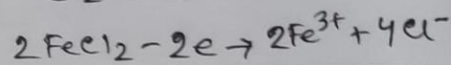
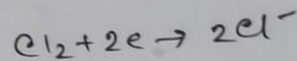
(i) x 4 then add (i) + (ii)



(i) x 3 then add (i) + (ii)



(ii) x 2 + (i) + (ii)



If k_1 & k_2 are the values of rate constant at temperature T_1 & T_2 we can write,

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[\frac{T_2 - T_1}{T_2 T_1} \right]$$

Half life of a Reaction: Half-life ($t_{1/2}$) in chemistry is the time required for half of a reactant to be consumed in a chemical reaction.

1. Zero-Order Reaction: $t_{1/2} = \frac{[A]_0}{2k}$
2. First-Order Reaction: $t_{1/2} = \frac{0.693}{k}$
3. Second-Order Reaction: $t_{1/2} = \frac{1}{k[A]_0}$

4) The integrated rate equation for a first order reaction, $k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$

$$\text{or } k = \frac{\ln 2}{t} \quad \text{or } k = \frac{1}{t} \ln \frac{[A]_0}{[A]}$$

half-life: $t_{1/2} = \frac{\ln 2}{k}$

5 Math: for first order reaction,

$$t_{1/2} = \frac{0.693}{k}$$

$$\Rightarrow k = \frac{0.693}{t_{1/2}} = \frac{0.693}{100} \\ = 0.00693 \text{ s}^{-1}$$

When 75% = $\frac{3}{4}$ initial concentration has reacted,
it is reduced to $\frac{1}{4}$.

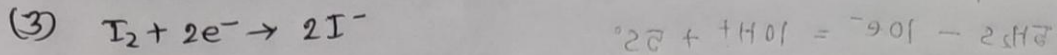
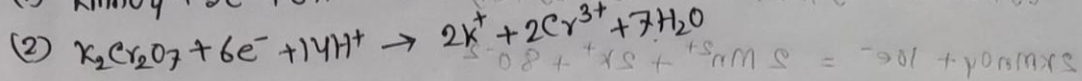
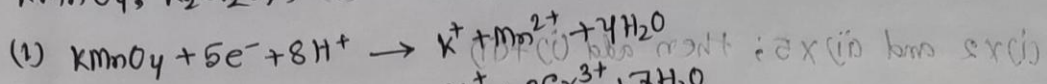
Substituting values in the rate equation

$$t_{3/4} = \frac{1}{k} \ln \frac{[A]_0}{\frac{1}{4}[A]_0} \\ = \frac{1}{0.00693} \ln (4) \\ = 200 \text{ sec (Ans.)}$$

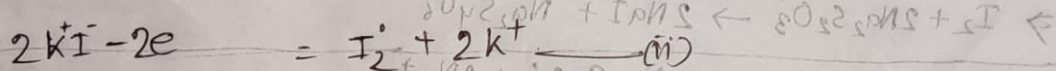
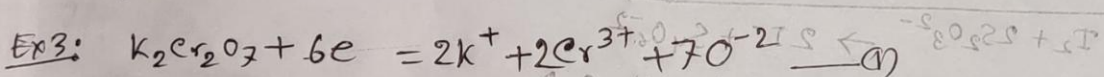
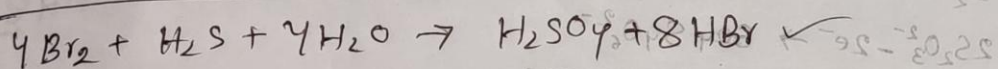
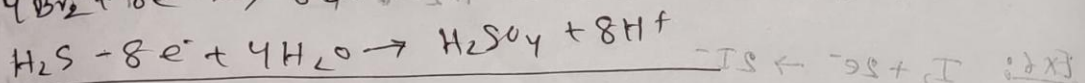
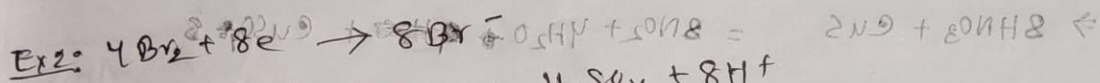
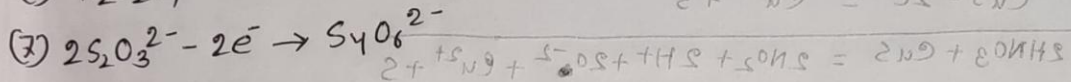
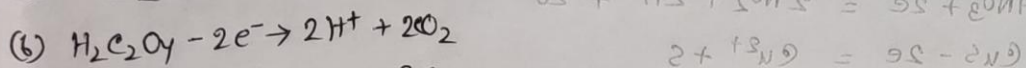
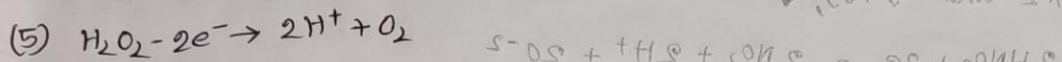
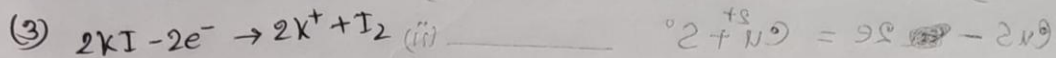
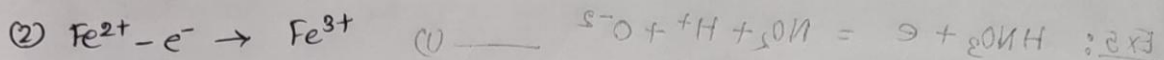
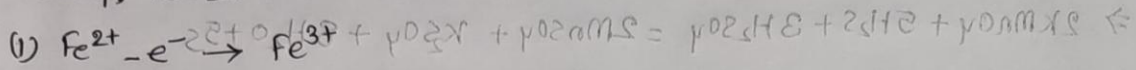
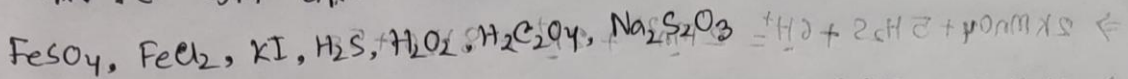
~~K = 0.00693~~

$$6. \quad t_{1/2} = \frac{0.693}{k} \\ = \frac{0.693}{0.450} = 1.54 \text{ sec}$$

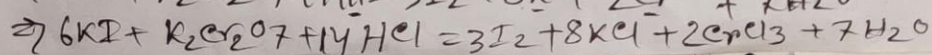
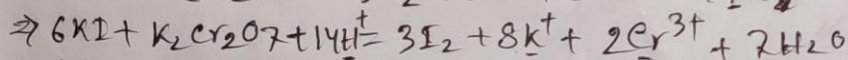
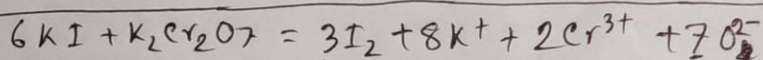
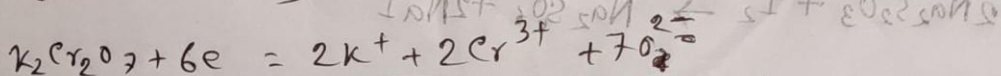
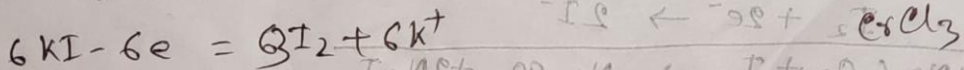
জারক e^- গ্রহণ বিজারক অর্ধ বিক্রিয়া : $M^{+n} + e^- = M^{(n-1)+}$: (i)
 $KMnO_4, K_2Cr_2O_7, I_2$ $0 + 1H^+ = -95 - 25H$

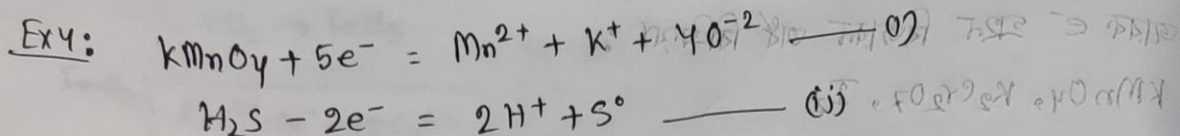


বিজারক : e^- ত্যাগ জারক অর্ধ বিক্রিয়া : $M^{(n-1)+} = M^{+n} + e^-$: (ii)

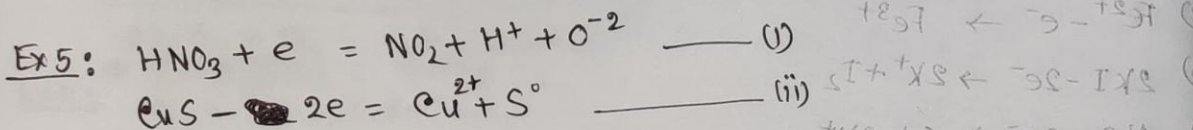
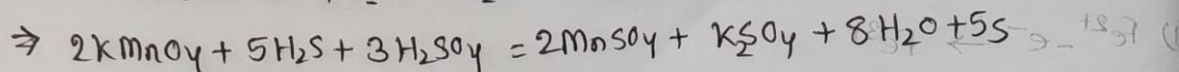
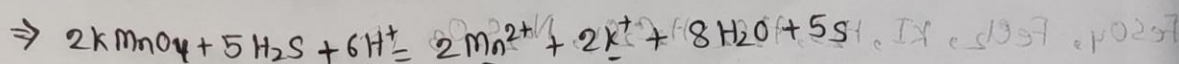
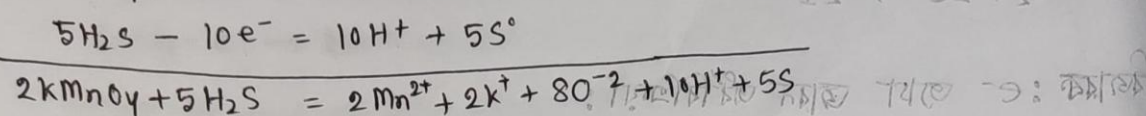
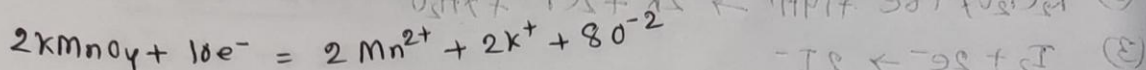


(ii) x3 and (i) + (ii)

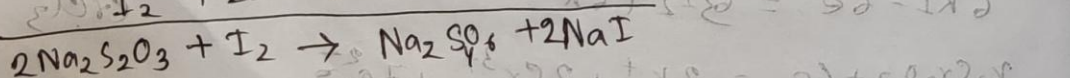
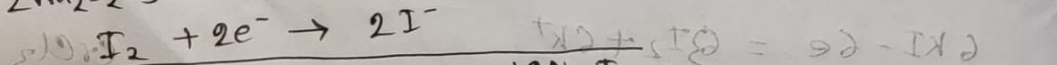
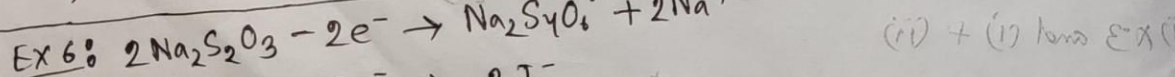
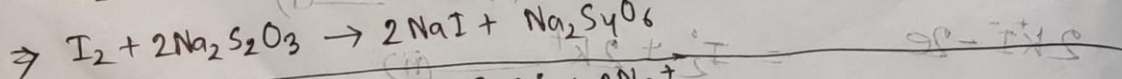
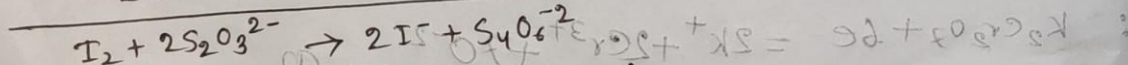
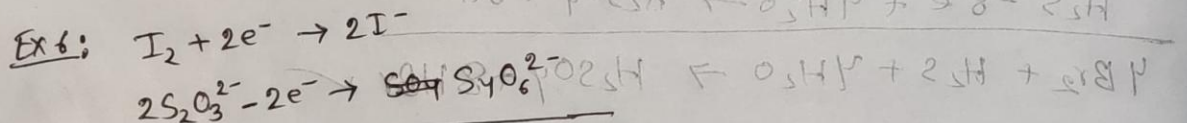
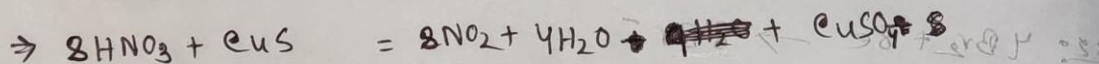
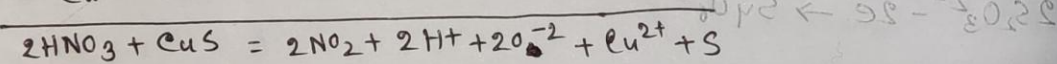
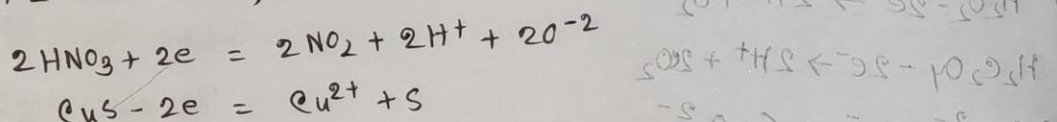


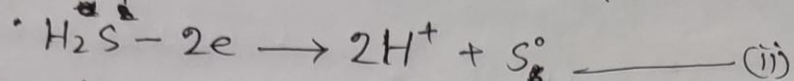
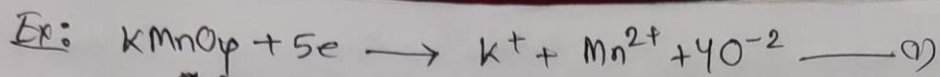


(i) $\times 2$ and (ii) $\times 5$; then add (i) + (ii)

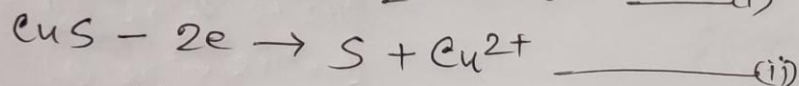
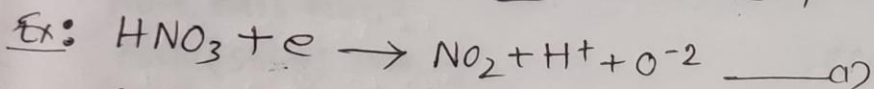
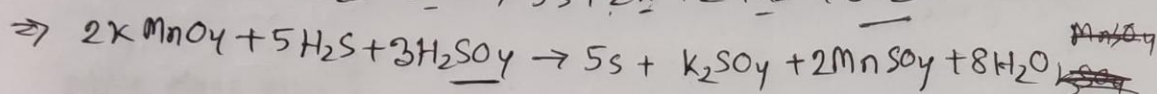
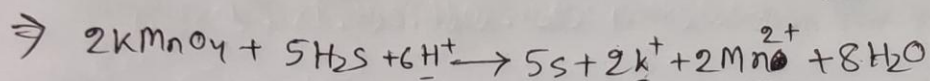
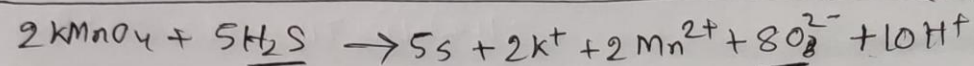
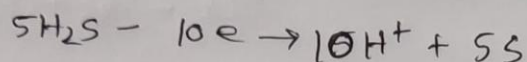
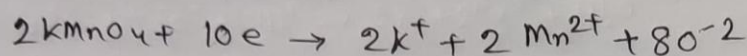


(i) $\times 2$ and (i) + (ii),





(i) x 2 and (ii) x 5 then add,



(i) x 2 and (ii) x 1

